

Exponent Laws with Variable Bases

Question Bank 3.T1

Name _____

Date _____

Period _____

Show your work. Circle your final answer.

For questions 1 through 6, write an equivalent expression as a single monomial. Assume no variable equals 0.

1. $x^4 \cdot x^9$

2. $(n^5)^6$

3. $(3y^4)^3$

4. $\frac{a^9}{a^3}$

5. $(-6x^2)(2x^7)$

6. $\frac{20p^5q^3}{5p^2q}$

For questions 7 and 8, apply more than one law. Write each answer as a single monomial.

7. Write an equivalent expression.

a. $(2x^3y)(5x^2y^4)$

b. $(4m^2n^3)^2$

c. $(a^2b)^3 \cdot ab^4$

8. Write an equivalent expression.

a. $\frac{(3m^2n)^3}{9m^4n}$

b. $\frac{x^8y^5}{x^8y^2}$

c. $\frac{(2k^4)^3}{4k^7}$

Working backwards.

9. Find the value of n that makes each statement true.

a. $x^3 \cdot x^n = x^{11}$

b. $(y^n)^4 = y^{20}$

c. $\frac{m^9}{m^n} = m^2$

10. A rectangle has length $6x^4$ and width $2x^3$.

a. Write its area as a single monomial.

b. A square has side length $4y^5$. Write its area as a single monomial.

11. Which expression is equivalent to $(5m^3)^2$?

- ☐ A. $5m^6$
- ☐ B. $25m^5$
- ☐ C. $10m^6$
- ☐ D. $25m^6$

12. Select ALL of the expressions that are equivalent to x^{12} .

☐ A. $(x^4)^3$

☐ B. $x^2 \cdot x^6$

☐ C. $(x^6)^2$

☐ D. $\frac{x^{15}}{x^3}$

☐ E. $x^4 \cdot x^3$

☐ F. $x^4 \cdot x^8$